

# Algebra II with Trigonometry

## Chapter 5.3

### Mental Math (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:  
[4401269\\_alg-2trig-h-chp-5-3-note-docx.pdf](#)

**Sine from Coordinates** If the point  $(3, 4)$  is on the terminal side of an angle in standard position, what is the value of the sine of that angle?

**Quadrant Signs** In which quadrant must an angle lie if its tangent is negative and its sine is also negative?

**Reference Angle Calculation** What is the reference angle for an angle measuring 300 degrees?

**Pythagorean Identity** If the sine squared of an angle is 0.2, what is the cosine squared of that same angle?

**Triangle Area** A triangle has two sides of length 10 and 12, and the included angle between them is 30 degrees. What is the exact area of the triangle?

**Reciprocal Functions** If the cosine of an angle is  $\frac{2}{7}$ , what is the exact value of the secant of that angle?

**Evaluating Cosine** What is the exact value of the cosine of 120 degrees?

**Cosine from Sine** If the sine of an angle is  $\frac{3}{5}$  and the angle lies in Quadrant II, what is the exact value of its cosine?

**Undefined Tangent** Which two angles between 0 degrees and 360 degrees (inclusive) have an undefined tangent?

**Tangent to Secant Identity** If the tangent squared of an angle is 8, what is the secant squared of that same angle?